MODULE 1

What Is LiDAR?

SIMPLE APPROACH

Close your eyes and clap your hands in a big gym. You hear an echo. If the wall is far away, the echo takes longer to come back. If it is close, the echo comes back fast. A bat flies in the dark this way!
A LiDAR does the same thing — but instead of a clap it uses a tiny flash of laser light, and instead of ears it has a super-fast light catcher. It does this hundreds of thousands of times every second, in every direction, so it can "see" the shape of everything around it: roads, trees, buildings, people. Each echo becomes one dot. Millions of dots together make a 3-D picture called a **point cloud**.

1.1 The name

LiDAR stands for **Li**ght **D**etection **A**nd **R**anging. "Ranging" is the surveyor's word for measuring distance. A LiDAR is, at heart, an electronic distance meter (EDM) — the same physics as the EDM inside a total station — that fires extremely fast and sweeps its beam in many directions.

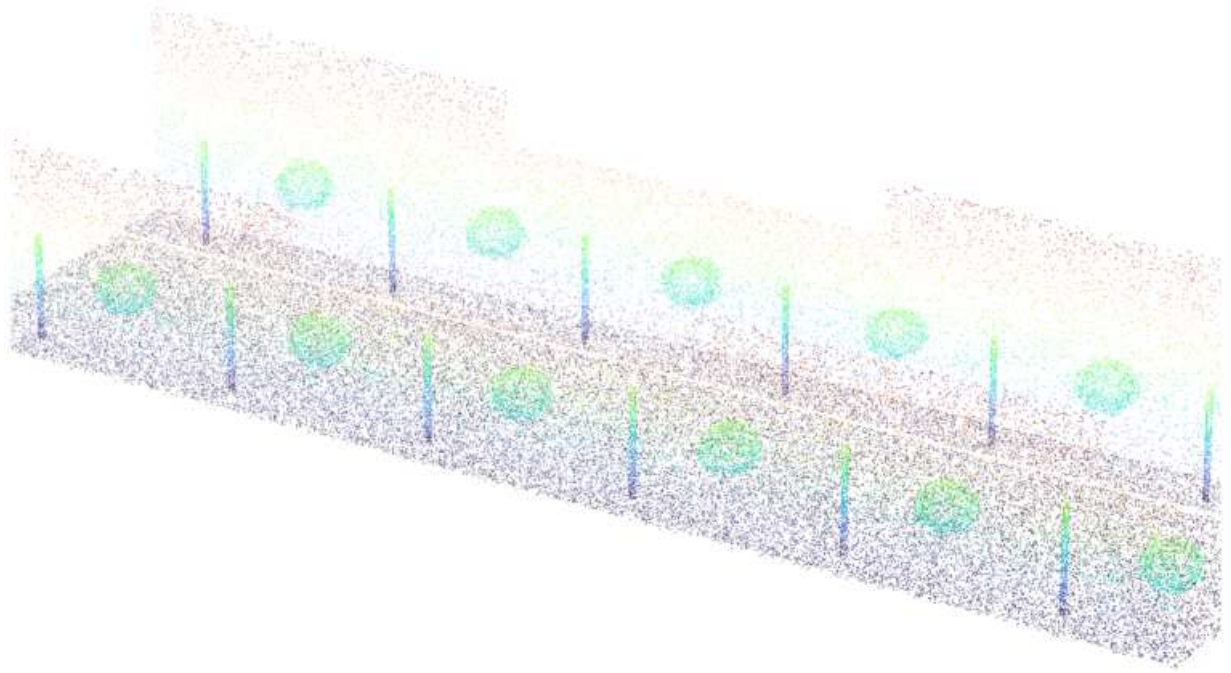


Figure 1.1: A synthetic mobile-LiDAR point cloud of a street, colored by height: road, curbs, poles, trees and building faces. Every dot is one laser echo.

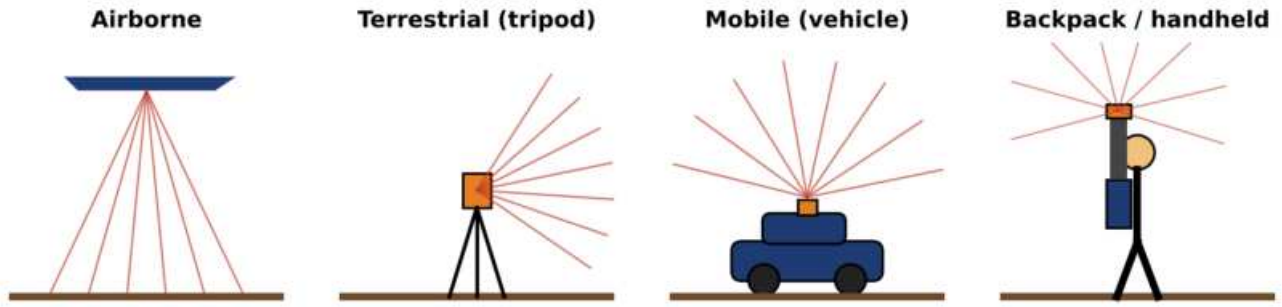


Figure 1.2: The same measuring idea mounted four ways.

1.2 The four families of LiDAR

Platform	How it moves	Where the position comes from	Typical use
Airborne	aircraft / drone	GNSS + IMU (very open sky)	terrain, floodplains
Terrestrial (TLS)	stands still on a tripod	surveyed setups, targets	buildings, plants, forensics
Mobile (MLS)	car, train, boat	GNSS + IMU + LiDAR SLAM	corridors, roads, rail
Backpack / handheld	a walking person	IMU + LiDAR SLAM (+ control)	indoors, tunnels, under canopy

This course concentrates on the two moving families at the bottom of the table: **mobile** and **handheld** systems. They are the hardest, because the instrument never holds still — and the most productive, because you can capture a kilometer of street in the time it takes to set up a total station a few times.

1.3 The one big problem: the instrument is moving

BIG IDEA

A laser can only tell you *"that wall is 25.000 m away in that direction from me."* If you do not know exactly where "me" is — and which way "me" was pointing — at the *exact instant* of each pulse, the measurement is useless. Most of this course is about knowing "me": the sensor's **pose**.

ENGINEER'S VIEW

For a moving scanner each laser return must be placed in the project frame using the full six-degree-of-freedom pose (three positions X, Y, Z and three angles roll, pitch, yaw) of the sensor at the time that particular pulse was fired. A spinning scanner at 10 Hz fires up to a few million pulses per second, so the pose must be known at very high rate. That is achieved by combining:

1. an **inertial measurement unit (IMU)** for high-rate, short-term motion (Module 7);
2. **precise time synchronization** so every measurement shares one clock (Module 8);



Figure 1.3: Left: the same scans placed with a bad trajectory form the famous "bird's nest." Right: with a correct high-rate trajectory every scan stacks into a crisp map.

3. measured **lever arm and boresight** between the sensors (Module 9);
4. a **trajectory estimator** — GNSS/INS fusion and/or LiDAR SLAM — that keeps drift bounded (Modules 10–12);
5. **ground control and independent checkpoints** that tie the result to a datum and prove its accuracy (Module 13).

1.4 Why surveyors already understand most of this

Mobile-LiDAR idea	Classical surveying equivalent
Time-of-flight range	EDM distance
Polar-to-Cartesian conversion	Total-station angle + distance → coordinates
Pose (position + orientation)	Instrument station + backsight orientation
IMU dead reckoning	An open traverse — error accumulates
Loop closure	Closing a traverse
Factor-graph optimization	Least-squares network adjustment
Control points vs checkpoints	Constrained points vs independent check shots
Error budget (root-sum-square)	Traverse / level-network design

1.5 What you will be able to do at the end

- Compute a range from a time of flight and explain why we divide by two.
- Turn a range and two angles into X, Y, Z — and then into project coordinates using lever arm, boresight, attitude and trajectory.
- Predict how fast an IMU of a given grade drifts, and how far a timing or boresight error moves a point.
- Explain, with numbers, what an extended Kalman filter, ICP, and a factor graph actually do.
- Read the LIO-SAM paper's equations and know what every symbol means.
- Build an error budget, compute RMSE from checkpoints, and report accuracy in the language of current standards.

TRY IT YOURSELF

1.1 (Simple) If you clap and hear the echo after 1 second, and sound travels about 343 meters each second, how far away is the wall? (Hint: the sound went there *and* back.)
1.2 (Engineer) List the five ingredients (besides the laser) needed to turn a moving scanner's ranges into a survey-grade map.

Answers: worked solutions are in Appendix A, "Answers to Try It Yourself," of the Reference Book, starting on page 158.

SOURCES FOR THIS MODULE

Shan, T. *et al.* (2020), LIO-SAM, IEEE/RSJ IROS, arXiv:2007.00258; *Mobile LiDAR Basics for Surveying and Mapping* seminar (2026); *LiDAR Mobile Mapping with SLAM* course outline (2026).